

# Cell Biology and Biochemistry

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**Science**

**May, 2026**

## Cell Biology and Biochemistry (A07896)

**Short Title:** Cell Biology and Biochemistry  
**Faculty** Science and Computing  
**Department** Science  
**Cluster** Biology  
**Author** ROMA HONY  
**Credits** 5

**Level:** Introductory

### Description of Module / Aims

The aim of this module is to develop an understanding of the fundamentals of cell biology and biochemistry.

### Programmes

|           |                                                                             |           |
|-----------|-----------------------------------------------------------------------------|-----------|
| BIOL-0002 | BSc (Common Entry) (WD_SSCIE_D)                                             | 1 / 2 / M |
| BIOL-0002 | BSc (Hons) in Agricultural Science (WD_SAGRI_B)                             | 1 / 2 / M |
| BIOL-0002 | BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)                      | 1 / 2 / M |
| BIOL-0002 | BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B) | 1 / 2 / M |
| BIOL-0002 | BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)                           | 1 / 2 / M |
| BIOL-0002 | BSc (Hons) in Physics for Modern Technology (WD_KPHE_B)                     | 1 / 2 / M |
| BIOL-0002 | BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)                           | 1 / 2 / M |
| BIOL-0002 | BSc in Agricultural Science (WD_SAGRI_D)                                    | 1 / 2 / M |
| BIOL-0002 | BSc in Food Science with Business (WD_SFDSC_D)                              | 1 / 2 / M |
| BIOL-0002 | BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)        | 1 / 2 / M |
| BIOL-0002 | BSc in Pharmaceutical Science (WD_SPHSC_D)                                  | 1 / 2 / M |

### Indicative Content

- Structure, properties, occurrence and roles of biological macro-molecules
- The ultrastructure of prokaryotic and eukaryotic cells and the functions and interrelation of various cell organelles
- The structure and function of cell membranes (transport, cell signalling/communication)
- Enzymatic structure and factors affecting enzyme activity
- An introduction to microbial metabolism

### Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Distinguish between the four classes of biological macromolecules.
2. Illustrate the structural features of prokaryotic and eukaryotic cells and explain their functions.
3. Outline the importance of enzymes and the various factors affecting enzyme activity.
4. Complete and examine qualitative and quantitative biochemical assays.
5. Construct and interpret graphs using experimental data.

### Learning and Teaching Methods

- Cell biology and biochemistry is delivered in a multi-modal approach.
- The module consists of lectures, laboratory practicals and online tutorials (on or off site).

- The teaching strategy is to adopt a blended learning approach.
- Lecture sessions are delivered using a flipped classroom style.
- Resource material is made available to the student prior to the lecture session.
- The student completes e-activities in the form of quizzes which are designed pre and post lecture sessions.
- Audio and video files of module content are provided to cater for the variety of learner types (class size < 50).
- Laboratory practicals and tutorial sessions are linked.
- Pre-practical quizzes are used as a formative learning method and these can be completed during tutorial sessions on or off-site.
- The entire above are delivered through Moodle using blended and e-learning.

## Assessment Methods

|                           | Weighing | Outcomes Assessed |
|---------------------------|----------|-------------------|
| Final Written Examination | 50%      | 1,2,3,5           |
| Continuous Assessment     | 50%      | 4,5               |
| <i>Practical</i>          | 50%      | 4,5               |

## Assessment Criteria

**<40%:** No understanding of basic cell biology and biochemistry. Unable to carry out critical thinking or practical skills.

**40%-49%:** Demonstrate a limited knowledge of basic cell biology and biochemistry. May have some difficulty with critical thinking and problem solving. Adequate practical skills.

**50%-59%:** A good understanding of cell biology and biochemistry with demonstration of some critical thinking and problem solving skills. Good practical skills.

**60%-69%:** In addition to the above, good critical thinking skills and showing a high level of competence and efficiency in theoretical and practical components.

**70%-100%:** A highly developed and extensive understanding of cell biology and biochemistry. Excellent technical ability in carrying out and reporting on practical tasks.

### Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 36        |           |
| Practical            | 24        |           |
| Independent Learning | 75        |           |

## Essential Material

- "Cell Biology and Biochemistry module Moodle page." <http://moodle.wit.ie>
- Madigan, M.T., J.M. Martinko, D.H. Buckley, D.A. Stahl and T. Brock. *Brock Biology of Microorganisms*. 14th ed. Essex, UK: Pearson Education Ltd., 2015.

## **Requested Resources**

- EQUIPMENT: Blackboard
- Room Type: Biology Lab
- Lecture Room: Loose Seated
- EQUIPMENT: MAC PCs